



EXAM PAPERS PRACTICE

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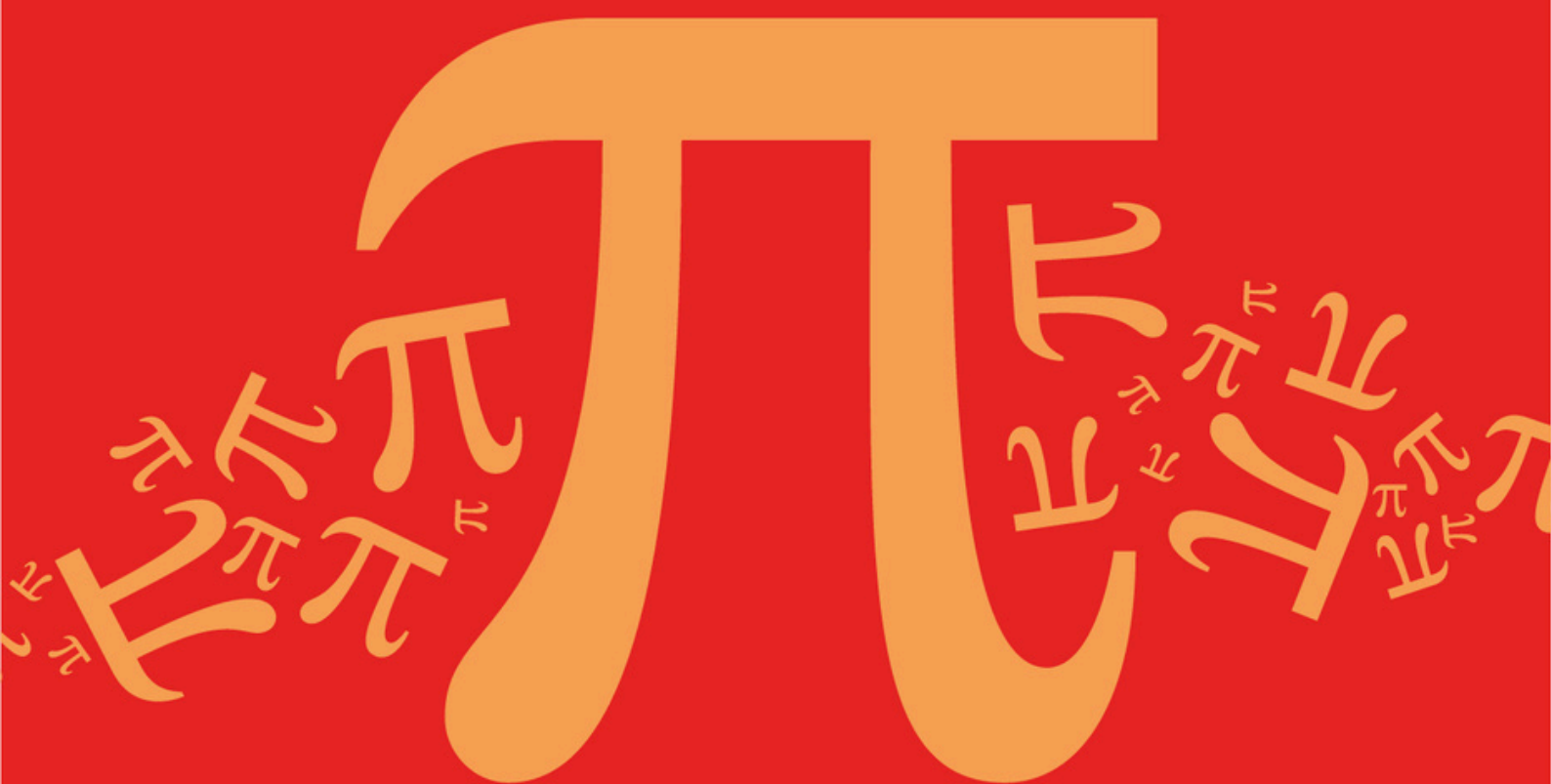
Practice questions created by actual
examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and
thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DA: Graphs

Sub topic

DA6: Simple Graphs and Trees

Mark Scheme

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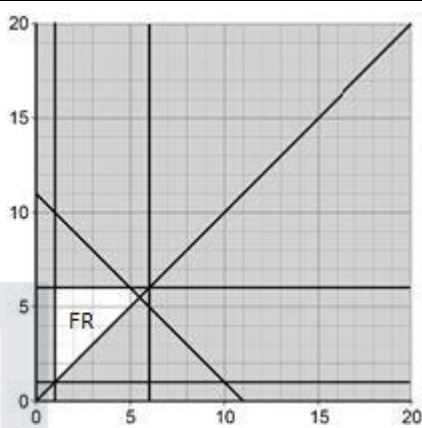
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

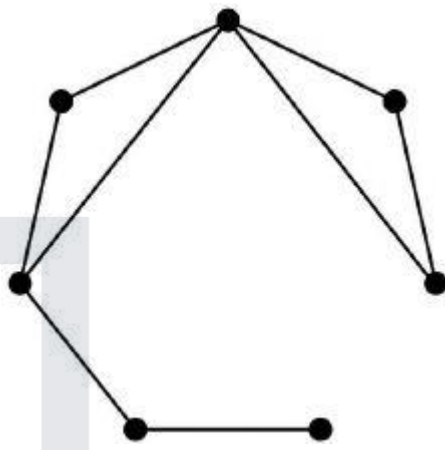
Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Draws $y = x$	AO1.1a	M1	
	Clearly labels (or shows by shading) the correct feasible region	AO1.1b	A1	
(ii)	Calculates the correct value of $5x + 4y$ at $(5, 6)$ or $(5.5, 5.5)$	AO1.1a	M1	Value at $(5, 6) = 49$ Value at $(5.5, 5.5) = 49.5$
	Finds the correct maximum value of $5x + 4y$	AO1.1b	A1	Maximum value is 49.5
(b) (i)	Explains the implication of the graph being connected Must refer explicitly to 'connected'	AO1.2	B1	The graph is connected, so no vertex can have degree 0.
	Explains the implication of the graph being simple Must refer explicitly to 'simple'	AO1.2	B1	The graph is simple, so each vertex can only connect at most once to each of the 6 other vertices.
(iii)	Explains why x and y cannot both be 6	AO2.4	E1	If x and y are both 6 then there would be two vertices that are both adjacent to the other five vertices.
	Completes rigorous argument for $x + y \leq 11$	AO2.1	R1	However, this would mean that there would be no vertex of degree 1 so one of x or y has to be at most 5. Hence $x + y \leq 5 + 6$
(iv)	Identifies the need for integer solutions	AO3.5b	B1	x and y must both be integers

	Marking Instructions	AO	Marks	Typical Solution
(c) (i)	Uses degree sum $= 2 \times 8$ to deduce $x + y$	AO2.2a	M1	$1 + 2 + 3 + v + w + x + y = 2 \times 8$ $16 - (1 + 2 + 3 + 4) = x + y$ $x + y = 6$

	Infers there are multiple solutions and finds two correct feasible pairs of x and y values	AO2.2b	M1	$x = 1, y = 5;$ $x = 2, y = 4;$ $x = 3, y = 3.$
	Finds all three pairs and no others Condone pairs written as coordinates	AO1.1b	A1	
(ii)	Starts to show configuration of G by drawing a connected graph with exactly 8 edges OR by listing/labelling the vertex degrees: 1, 2, 2, 2, 2, 3, 4	AO3.1	M1	
	Draws a correct graph with degrees: 1, 2, 2, 2, 2, 3, 4 NB Multiple correct solutions, check by counting degrees	AO3.2a	A1	
Total 14 marks				

Q2.

(a) (i)

AB	6.1
BC	7.4
BE	9.7
DE	7.2
EF	10.6
EH	12.5
HI	6.7
GH	8.9

Prim's, 1st 3 correct, must be edges not lengths and no cycles

8 edges

EF 5th

M1

B1

A1

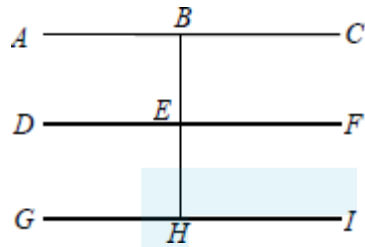
All correct

A1

(ii) (Length =) 69.1

B1

(iii)



Spanning tree with 9 vertices and 8 edges

All correct, including labelling

M1

A1

7

(b) (i) GH

B1

(ii) EF

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B1

2

(c) (i) 1st AB

B1

(ii) Last EH

B1

2

[11]

Q3.

(a) (i)

EG	$\left(\begin{array}{c} 2.3 \\ 2.5 \\ 2.9 \\ 3.1 \\ 3.2 \\ 3.4 \\ 3.6 \\ 3.9 \\ 5.4 \end{array} \right)$
AB	
IJ	
AC	
AD	
HJ	
GJ	
BE	
FI	

SCA, Kruskal's, 1st 3 edges correct, must be edges not lengths, and no cycle in solution

9 edges

AD 5th

All correct

M1

B1

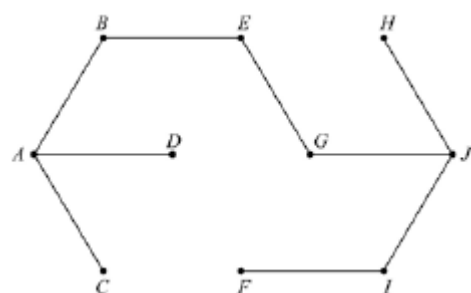
A1

A1

(ii) 30.3

B1

(iii)



Spanning tree with 10 vertices and 9 edges.

M1

All correct including labelling

A1

7

(b) (i) FI

B1

(ii) DA

B1

2

[9]

Q4.

(a) (i)

AD	4
AB	6
AC	16
DE	19
EG	10
GI	12
IH	13
IF	17

Using Prim's, first 3 edges correct, 6+ edges, no cycles, must have edges not lengths

8 edges

GI 6th

All correct

M1

B1

A1

A1

4

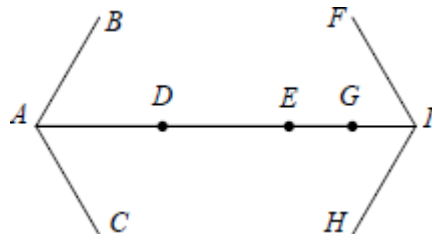
(ii) 97

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B1

1

(iii)



ST with 6+ edges

All correct including labels

M1

A1

2

(b) (i) IF

B1

1

(ii) AC

B1

1

[9]

Q5.

(a) Max 5

B1

Min 1

Do not allow 1° or 5°

B1

2

(b) $4x - 12 \geq 1$ (or > 0)

$$\left(x \leq \frac{13}{4} \right)$$

Or

$4x - 12 \leq 5$ (or < 6)

$$\left(x \leq \frac{17}{4} \right)$$

Or

$2x - 4 \leq 5$ (or < 6)

$$x \leq \frac{9}{2}$$

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Any one of these inequalities

OR

Exhaustive check of all values from 1 to 5 inclusive, condone one omission.

M1

$x = 4$

First inequality and one of the other two, or completely correct exhaustive check, and $x = 4$

A1

2

Alternative solution

Sum of degrees = $11x - 24$ must be even

$\Rightarrow x$ is even

$x - 2 > 0 \Rightarrow x > 2$

M1

$x \leq 5$

Hence $x = 4$

A1

[4]

Q6.

(a) Min MST

4 edges

M1

$$= 8 + 10 + 10 + 11 = 39$$

A1

2

(b) Max MST = $8 + 17 + 17 + 18$
 $8 + 18 + 2$ others

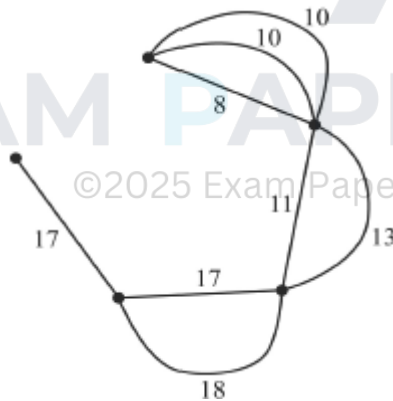
M1

$$= 60$$

A1

2

(c)



Connected graph with 5 vertices (all edges numbered, from G)

M1

MST = 53
8, 11, 17, 17 or 8, 10, 17, 18

A1

other edges OE

A1

(other possibilities not shown)
(all edges numbered, from G)

3

[7]

Q7.

(a) (i) 9

B1

1

(ii) $n - 1$

B1

1

(b) (i) GI 5

9 edges

AB 6

SCA

EI 7

start with GI

B1

M1

A1

BD 8

~~EC~~ 9

IJ 10

IJ fifth

A1

HJ 11

~~HI~~ 12

111 12,

AF 13

DE 14

CG 15

all correct

A1

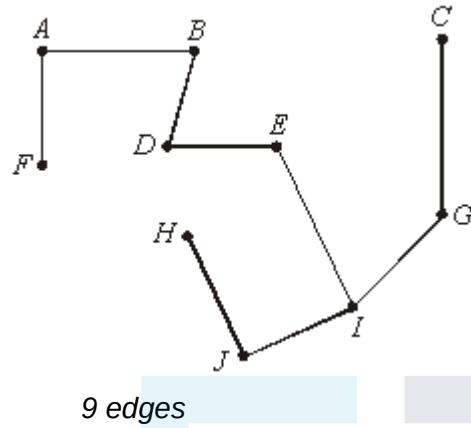
5

(ii) 89

B1

1

(iii)



M1

A1

2

[10]

Q8.

(a) $\text{Min} = 4 + 7 + 7 + 7$
 $= 25$

B1

1

(b) $\text{Min}(H) = 4 + 7 + 7 + 7 + 8$
 $= 33$

B1

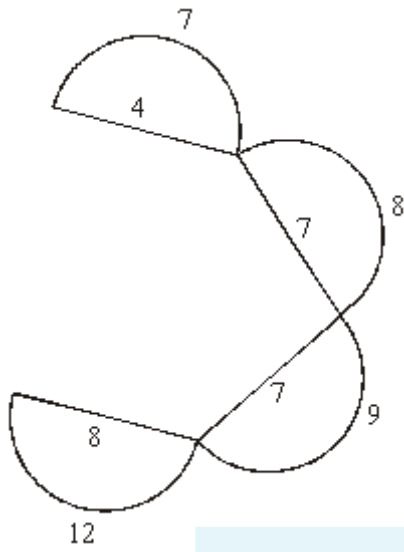
1

(c) $\text{Min}(E) = \Sigma = 62$

B1

1

(d)



5 vertices

8 edges

All correct

M1

m1

A1

3

[6]

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Q9.

(a) (i) 4

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B1

1

(ii) 2

B1

1

(iii) 5

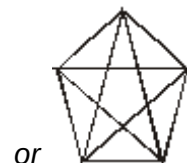
B1

1

(b)



or equivalent



or

B1
1
[4]

Q10.

- (a) (i) $(ABDK) = 3$

B1
1

- (ii) $MST = (n - 1)$

M1

$= 15$

A1
2

- (b) (i) Odd vertices A and P
Route length = all edges + repeated edges (answer > 30)

M1 M1

Route length = all + (3)

A1

$(33 + 3) = 36$

B1
4

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- (ii) Repeated: AE, EL, LP

B1
1

- (iii) E and L now odd

M1

Their $(33) - 2 + 1 = 32$

A1F
2

[10]

Q11.

- (a) (i) Odd vertices ($\Rightarrow$ Repeats)
Eularian not enough

E1
1

- (ii) Odd $BCDE$
considering odds

M1

$$\begin{aligned}
 BC + DE &= 14 \\
 BD + CE &= 12.5 + 10.5 = 23 \\
 \underline{BE} + \underline{CD} &= 9 + 4.5 = \underline{13.5}
 \end{aligned}$$

A2,1,0

$$\begin{aligned}
 \text{Distance} &= 13.5 + 82 \\
 &= 95.5 \\
 &82 + \text{their (mir)}
 \end{aligned}$$

M1

A1F

Route: $A B C D E F B E B D C A$
or equivalent

B1

6

(b) (i) $5 \times 3 = 15$

B1

1

(ii) $(n-1)(n-3)(n-5)\dots \times 1$

M1

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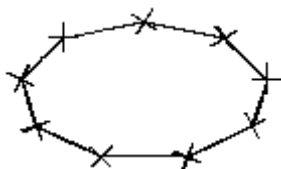
A1

2

[10]

Q12.

(a)



or equivalent

M1A1

2

(b) (i) $d = 2, 4, 6, 8$

M1A1

2

(ii) Edges = (9), 18, 27, 36

$$9 \frac{d}{42}$$

$$- 1 EE$$

M1

A2F

3

[7]

Q13.

(a) 1

B1

1

(b) 2

B1

e.g. *AB* and *ED*

B1

2

(c) (i) e.g. add *AB*

B1

Eulerian trail *AEFABCDB*

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M1 A1

3

(ii) e.g. adding *AB* twice creates a connected graph with all degrees even

B1

1

(iii) 4

B1

e.g.



M1A1

3

[10]

Q14.

- (a) (i) Sum of degrees = $5d + 7$

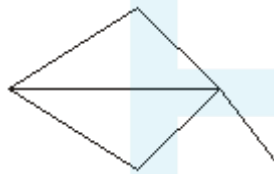
B1

This must be even and so d must be odd

B1

2

- (ii) $d \geq 3$ makes $d + 3$ too big, so $d = 1$:
e.g.



B1

M1A1

3

- (b) (i) each vertex is joined once to at most the other 9 vertices

B1

1

- (ii) If all the degrees were different they would be 0 1 2 3 ... 9.

M1

But that would give a vertex joined to none and another joined to them all – clearly impossible.

A1

2

[8]

Examiner reports

Q1.

This was a question which allowed students to demonstrate their understanding and application of mathematics in an unfamiliar situation, which is a key feature of the new specification. The majority of students scored 2/2 in part (a)(i). The most common error was to select an incorrect region as the feasible region. Some students did not draw the line $y = x$. Less than half of the students scored 2/2 in (a)(ii), with the most common mistakes being selecting the wrong vertex. The best solutions included an objective line which made the selection of the optimal vertex easier.

Nearly three quarters of students were able to explain the term 'connected' and relate it to the inequalities in (b)(i). However, less than half of students were able to correctly explain how the graph being simple related to the inequalities. Errors that were made included no direct reference to the degree of the vertices or the particular property of the graph that was being used.

Only 6% of students scored both marks in (b)(iii), with many students erroneously restating what they had written in (b)(ii). There were a significant number of no responses on this part. 10% of students scored the mark in part (b)(iv), with common errors being the adding together of two inequalities or the stating of a previous inequality.

Only a third of students scored any marks in (c)(i), with the vast majority not using the sum of the degrees being equal to twice the number of edges in the graph. Many answers did not write the values of x and y as pairs, and some solutions included pairs of x and y values that did not satisfy all of the inequalities.

More than half of students were able to score at least one mark in (c)(ii), with many students scoring 2/2 having not scored full marks on (c)(i). The better solutions included writing down the degree of each vertex next to the respective vertex. The loss of marks was most commonly due to having too few or too many edges on the graph.

Q2.

The three sections of part (a) were invariably correctly answered and clearly presented. Part (b) proved to be more challenging. Students were asked for edges but 'G' and then 'F' were quite common as answers. Part (c) was well answered with very few incorrect answers.

Q3.

Generally this was a very high scoring question with many students scoring full marks. The three parts of part (a) were usually shown clearly and calculated accurately. Many students are clearly taught to write down all the sides in order of ascending length and then to tick or cross them according as to whether they are to be included, without writing the resulting spanning tree out separately. Very few students forgot to label their diagram. In the second part, a few students confused the request for edges with one for vertices.

Q4.

The question was almost always very well answered. In part (b) weaker students often confused 'edge' and 'vertex'.

Q5.

Very few marks were scored. Around half the candidates appeared to think this question was about hexagons and certainly about the sizes of angles in degrees. From those who correctly interpreted the question the answers 2 and 6 to part (a) were about as common as the correct answers 1 and 5.

Most scoring attempts at part (b) were by numerical checking rather than the use of algebraic inequalities and revealed that most candidates had little or no concept of 'show', let alone 'proof' in this question.

Q6.

The usual mark scored was 4 out of 7, with very few candidates getting part (b) or the final mark in part (c) due to adding in extra edges and changing the MST.

Part (a) was usually correct.

Part (b) was usually incorrect, with $13 + 17 + 17 + 18 = 65$ being the almost universal answer. The very few who answered correctly were by no means always drawn from those scoring highest on the rest of the paper.

In part (c), most of the candidature spotted one of the two correct MSTs. A number of candidates elected to draw only a correct MST. Those candidates who attempted to complete the diagram frequently forfeited marks as their extra edges impacted on their MST. A few candidates drew separate diagrams.

Q7.

This question was a good source of marks for virtually all candidates, with few scripts using the wrong algorithm. Candidates who listed all the edges in part (b) were successful, although a common mistake was including edge FH. Some diagrams of minimum spanning trees were not labelled at the vertices.

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Q8.

This question proved to be a good discriminator, requiring candidates to think of an abstract network. Part (a) was well answered, but correct answers to parts (b) and (c) were rarely seen. Candidates started part (d) well by drawing a network with 5 vertices and a minimum spanning tree of 26 units, but often went astray when adding the extra arcs, ending up with a graph for which the minimum spanning tree had a length of 25 units.

Q9.

Candidates generally scored high marks on this question, although full marks were rarely seen. Many candidates gave the wrong answer of 72 to part (a)(ii).

Q10.

Part (a) was well answered.

Part (b) proved to be more challenging. Candidates were able to make a good attempt at part (i), although a significant number used $16 + 2$ or 3. In part (ii), a common mistake was to omit the edge EL . Part (iii) proved to be a good discriminator. Candidates failed to realise that by removing AE and LP , then E and L would become the odd vertices.

Q11.

Although candidates realised that an optimal ‘Chinese postman’ route depended on odd vertices, there was a disappointing number who failed to pair off the odd nodes correctly to arrive at the correct minimum repetition. A surprising number of candidates who did arrive at the correct numerical answer failed to write down a route. Part (b) proved to be the most difficult part of the paper with only a few candidates obtaining the correct answer for part (b)(ii).

Q12.

This question proved to be the most challenging on the paper, with only a few candidates scoring full marks. As the question stated that the graph was simple and connected, then d could not exceed 9 and had to be even, but there were many scripts that had odd values of d or values that implied the graph was not simple. Part (b)(ii) was a standard result but was poorly answered with some candidates quoting non-integer values for the number of edges.

